

Where Should You Work?

A Data-Driven Location Choice Tool for Early-Career Workers

Name: Xiaojia Guo, Chenxi Zhang

1. Question

One of the most important decisions facing recent graduates and early-career workers is where to live and work. When comparing job opportunities, people often focus on salary because it is the most visible part of a job offer. However, a higher salary does not necessarily translate into a better standard of living. Housing costs, overall prices, and labor market conditions vary widely across states, meaning that the value of the same salary can differ substantially depending on location.

This project develops an interactive dashboard that helps users compare U.S. states using a broader set of economic indicators. Rather than focusing only on nominal wages, the dashboard combines salary data with housing costs, regional price levels, and labor market conditions. By bringing these factors together, it provides a more complete picture of affordability and purchasing power across states.

The central question of this project is whether a higher salary always leads to a better place to live and work. The dashboard is designed to help users evaluate the trade-offs between income, cost of living, and employment opportunities when making location decisions.

The primary audience for this project includes recent graduates, early-career professionals, and international students who may have flexibility in choosing where to begin their careers. These groups often compare opportunities across states but may not fully account for differences in housing costs and regional price levels. The dashboard aims to provide a simple and accessible tool for making more informed location choices.

2. Data

The dashboard combines information from four publicly available datasets to capture different dimensions of economic opportunity and affordability across U.S. states.

Average salary data come from the Bureau of Labor Statistics (BLS) Occupational Employment and Wage Statistics (OEWS) program. At the same time, state unemployment rates are obtained from the BLS Local Area Unemployment Statistics (LAUS) program. Together, these variables provide information about both earnings and labor market conditions. Average salary reflects the income opportunities available in a state, while unemployment serves as a broad indicator of labor market strength. Although users may already have a job offer when comparing locations, labor market conditions remain important because they affect future career opportunities. States with lower unemployment rates generally provide stronger labor demand, greater job mobility, and a wider range of employment options. Including unemployment therefore allows the dashboard to capture not only current earnings potential but also the broader economic environment in which workers build their careers.

Housing affordability is measured using median gross rent data from the U.S. Census Bureau's 2024 American Community Survey (ACS) DP04 Selected Housing Characteristics dataset. Housing is often one of the largest monthly expenses for recent graduates and early-career workers. As a result, comparing wages alone can be misleading if housing costs vary substantially across locations. Including rent data allows users to assess how much of their income may be absorbed by housing expenses in different states.

To account for differences in overall living costs, the dashboard also incorporates Regional Price Parities (RPPs) from the Bureau of Economic Analysis (BEA). RPPs measure price-level differences across states relative to the national average, which is set to 100. States with values above 100 are generally more expensive than the national average, while states below 100 are relatively less expensive. Using RPPs makes it possible to compare purchasing power rather than simply comparing nominal salaries.

After selecting the four datasets, several data preparation steps were required before they could be combined into a single dashboard dataset. State names were standardized, and non-state territories such as Puerto Rico, Guam, and the U.S. Virgin Islands were removed to ensure consistency across sources. The datasets were then

merged by state to create a unified dataset containing salary, unemployment, housing costs, and price-level information for all 50 states. This integrated dataset serves as the foundation for the dashboard's affordability and purchasing power comparisons.

3. Dashboard

The dashboard is designed to help users compare economic conditions across all 50 U.S. states using a small set of indicators related to income, affordability, and labor market conditions. Users can select up to five states for comparison, adjust an expected salary level, and view results through interactive charts and tables. The goal is not to identify a single "best" state, but rather to illustrate the trade-offs among earnings, housing costs, and overall affordability.

The dashboard combines user inputs with state-level economic indicators, including average salary, median gross rent, unemployment rates, and Regional Price Parity (RPP) data. While users can adjust the salary slider to explore different income scenarios, the underlying state rankings and affordability measures are based on state-level economic data collected from the four datasets described previously. These calculations are performed automatically within the dashboard, allowing users to instantly compare states under different state selections and salary scenarios.

One of the main measures used in the dashboard is Real Salary, which adjusts average annual salary for differences in regional price levels. Real Salary is calculated by dividing a state's average salary by its Regional Price Parity index relative to the national average:

$$\text{Real Salary} = \text{Average Salary} \div (\text{Price Index} / 100)$$

Because the national average price level is set equal to 100, states with higher RPP values are generally more expensive places to live. Adjusting salaries using RPPs provides a measure of purchasing power and allows users to compare how far income goes across different states.

The dashboard also calculates Rent Burden, which measures the share of annual income required to pay median rent in a given state:

$$\text{Rent Burden} = (\text{Median Rent} \times 12) \div \text{Average Salary} \times 100$$

Higher values indicate that housing costs consume a larger share of income and therefore suggest lower housing affordability. This measure helps users evaluate whether higher salaries are sufficient to offset higher housing costs.

Finally, unemployment rates are included to provide information about broader labor market conditions. Although unemployment does not predict outcomes for any individual worker, it serves as a useful indicator of labor demand and economic opportunities within a state.

Together, these measures allow users to compare states based not only on earnings but also on affordability and labor market conditions.

4. Insights

Several important patterns become clear when comparing states using the dashboard. First, states with the highest average salaries do not always offer the highest purchasing power. Once differences in living costs are accounted for through the Regional Price Parity adjustment, the ranking of states can change substantially. This suggests that nominal salary alone may not provide an accurate picture of economic well-being across locations.

Second, housing affordability varies substantially across states. In many cases, higher salaries are accompanied by significantly higher rents, reducing the financial advantage associated with greater earnings. The Rent Burden measure illustrates how housing costs can absorb a large share of income, particularly in high-cost states. As a result, workers may experience very different standards of living even when nominal salaries appear similar.

Third, labor market conditions do not always align with affordability. Some states combine strong labor markets with relatively high housing costs and price levels, while others offer lower living costs but fewer employment opportunities. This suggests that choosing where to live and work often involves balancing affordability against potential career opportunities.

Overall, the dashboard demonstrates that location decisions involve important trade-offs. Higher salaries do not automatically lead to better economic outcomes, and no single state performs best across all dimensions. Instead, individuals must weigh earnings, housing costs,

purchasing power, and labor market conditions according to their own priorities and career goals.

5. Limitations

Several limitations should be considered when interpreting the dashboard results. First, all indicators are measured at the state level and therefore represent averages rather than local conditions. Economic circumstances can vary substantially within a state, particularly between large metropolitan areas and rural communities. As a result, the dashboard may not fully capture the economic realities experienced by individual workers.

Second, the dashboard focuses on a limited set of indicators and does not account for several factors that may influence affordability. For example, state income taxes, occupation-specific wages, transportation costs, household composition, and other living expenses are not included in the current version. These factors may affect an individual's actual financial situation and could lead to different location preferences.

Third, the four datasets are collected and released by different agencies. Although the most recent data available were used for each source, the indicators do not necessarily refer to exactly the same time period. This is a common challenge when combining information from multiple public datasets.

Despite these limitations, the dashboard provides a useful framework for comparing economic conditions across states. Future versions could be expanded by incorporating metropolitan-area or county-level data, additional measures of affordability, and occupation-specific wage information. These enhancements would allow for more detailed comparisons and help users make location decisions that better reflect their individual circumstances.